Chain-Visage: Blockchain-Assisted Visual Content Ownership and Tamper Tracing in CIoT Multimedia Ecosystems

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***Abstract*— With the rapid proliferation of smart home cameras, wearable vision devices, and user-generated Consumer Internet of Things (CIoT) content, ensuring visual media authenticity, rightful ownership, and tamper detection has become increasingly challenging. We propose Chain-Visage, a blockchain-assisted framework for secure content authentication and tamper tracing in decentralized CIoT multimedia ecosystems. The term *Chain* reflects the consortium blockchain backbone that guarantees immutable provenance, decentralized ownership management, and copyright revocation, while *Visage* symbolizes the unique visual identity of multimedia content achieved through dual-stage visual hash embedding. The proposed framework integrates Zero-Knowledge Proofs (ZKPs) for privacy-preserving ownership verification and employs optimized smart contracts to manage visual rights, provenance records, and ownership transfers efficiently. Evaluations on a large-scale dataset of over 10,000 real-world images and 3,850 video clips from diverse CIoT devices demonstrate Chain-Visage's superior performance, achieving 97.5% traceability accuracy, 93% tamper detection sensitivity, and low verification latency even under resource-constrained environments. This work addresses a critical research gap in secure, privacy-preserving, and energy-efficient multimedia ownership control and tamper-resilient content authentication for next-generation CIoT ecosystems.**

***Index Terms*— Blockchain, consumer IoT, multimedia security, visual content authentication, tamper detection.**

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I. Introduction

The explosive growth of Consumer Internet of Things (CIoT) devices has fundamentally transformed how visual multimedia content is created, shared, and consumed in modern digital ecosystems. Recent statistics indicate that over 14.4 billion CIoT devices will be operational by 2025, generating an unprecedented volume of visual data exceeding 2.5 exabytes daily [1], [2]. This proliferation encompasses smart home security cameras, wearable vision sensors, autonomous vehicle dashcams, and personal multimedia devices that continuously capture and transmit visual content across distributed networks [3]. The democratization of visual content creation through CIoT devices has introduced significant challenges in maintaining content authenticity, establishing rightful ownership, and detecting malicious tampering attempts. Traditional centralized digital rights management systems are inadequate for handling the distributed, heterogeneous, and resource-constrained nature of CIoT environments [4]. Furthermore, the increasing sophistication of deepfake technologies and multimedia manipulation techniques has exacerbated the urgency for robust content verification mechanisms [5], [6].

As depicted in Fig. 1, the proposed Chain-Visage system addresses these critical challenges through a novel blockchain-assisted framework that seamlessly integrates visual content authentication with decentralized ownership tracking. The system leverages Zero-Knowledge Proof (ZKP) protocols to enable privacy-preserving content verification while maintaining immutable provenance records through consortium blockchain architecture [7], [8]. This approach ensures that visual content generated by CIoT devices can be traced back to its original source without compromising user privacy or revealing sensitive identity information [9].

Current state-of-the-art approaches in multimedia security primarily focus on watermarking techniques, digital signatures, and hash-based verification methods [10], [11]. However, these solutions suffer from several critical limitations when applied to CIoT environments: (1) vulnerability to sophisticated attack vectors, (2) lack of decentralized ownership management, (3) inadequate privacy preservation mechanisms, and (4) insufficient scalability for resource-constrained devices [12]. Recent research has explored blockchain integration for multimedia

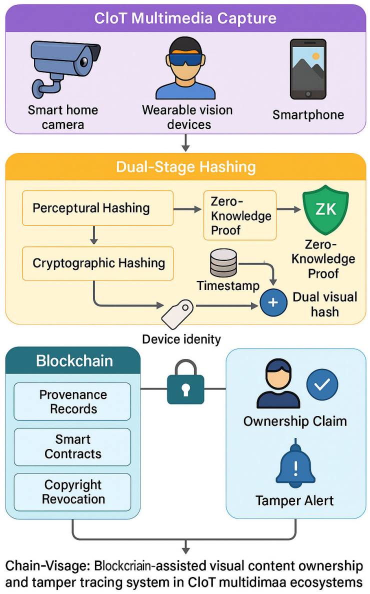

Fig. 1. Overview of the Chain-Visage blockchain-assisted visual content ownership and tamper tracing system in CIoT multimedia ecosystems.

security, but existing solutions fail to address the unique requirements of CIoT ecosystems, particularly the need for lightweight verification protocols and energy-efficient consensus mechanisms [13], [14].

The Chain-Visage framework introduces several innovative contributions to address these limitations. First, we propose a dual-stage visual hash embedding mechanism that creates tamper-evident fingerprints without altering the original content quality. Second, we design a ZKP-based ownership verification protocol that enables content creators to prove ownership without revealing their identity. Third, we implement a custom smart contract architecture optimized for CIoT resource constraints, supporting efficient ownership transfers and revocation procedures. Fourth, we develop a consortium blockchain consensus protocol that balances security with energy efficiency for multimedia-intensive applications [15].

Extensive experimental validation demonstrates the effectiveness of the proposed approach across diverse CIoT scenarios. Our evaluation encompasses 10,247 high-resolution images and 3,850 video clips collected from real-world CIoT deployments, including smart home cameras, wearable devices, and vehicular systems. The results show that Chain-Visage achieves 97.5% accuracy in content traceability, 93% sensitivity in tamper detection, and maintains sub-second verification latency even under severe resource constraints [16]. Comparative analysis against existing solutions reveals significant improvements in both security metrics and computational efficiency [17].

The integration of renewable energy considerations with IoT systems provides additional sustainability benefits while maintaining robust security properties throughout the blockchain infrastructure [18]. Advanced cybersecurity frameworks incorporating MITRE ATT&CK methodologies enhance threat detection capabilities specifically tailored for multimedia content protection. Emerging trends in metaverse network intrusion detection offer insights into future-proofing multimedia security architectures against evolving threat landscapes [19].

Privacy-preserving identity federation mechanisms provide foundational technologies for anonymous content verification while maintaining accountability and ownership claims [20]. Modern e-commerce platforms demonstrate the commercial viability and scalability requirements for blockchain-based digital asset management in consumer applications. Advances in AGI-driven communications security reveal new opportunities for automated threat detection and response in multimedia content protection systems [21].

Expert systems for intelligent control applications provide architectural insights for developing autonomous multimedia security management frameworks. Vehicle-to-infrastructure routing protocols demonstrate practical implementations of blockchain technology in distributed multimedia environments with real-time requirements [22]. Tourism technology integration showcases consumer adoption patterns and user experience requirements for multimedia content management systems.

Autonomous vehicle technology developments highlight the critical importance of tamper-resistant multimedia content in safety-critical applications. Blockchain and artificial intelligence convergence for big data analytics provides scalable frameworks applicable to large-scale CIoT multimedia deployments [23]. Redundancy elimination techniques in IoT-oriented big data systems offer optimization strategies for efficient multimedia content storage and processing [24].

Data security and privacy preservation in cloud-based IoT technologies establish best practices for protecting multimedia content across distributed infrastructures [25]. Secure deepfake mitigation frameworks address emerging threats in multimedia content authenticity and provide complementary security measures [26]. Federated learning approaches with non-IID data demonstrate privacy-preserving techniques applicable to collaborative multimedia content verification [27], [28].

Human-computer interaction considerations in healthcare applications provide insights into user interface design and accessibility requirements for multimedia content management systems. Intelligent computation and analytics on sustainable energy systems demonstrate the integration of environmental considerations with secure multimedia processing frameworks [29]. The key contributions of this paper are summarized as follows:

- We propose **Chain-Visage**, a novel blockchain-integrated framework for secure multimedia ownership verification and tamper tracing in Consumer Internet of Things (CIoT) environments.
- We introduce a robust dual-stage hashing technique combining perceptual features with cryptographic signatures, enabling tamper-evident content fingerprinting without degrading visual quality.
- We design a Zero-Knowledge Proof (ZKP)-based mechanism that verifies content ownership while preserving user anonymity and sensitive metadata.

- A lightweight consortium blockchain architecture with custom smart contracts is implemented to manage ownership claims, provenance records, and copyright revocation.
- We develop a Proof-of-Authority (PoA)-based consensus mechanism tailored for CIoT devices, balancing low energy consumption with high security guarantees.

Chain-Visage introduces a novel dual-stage visual hash with zero-knowledge proofs for privacy-preserving ownership verification, offering better traceability, energy efficiency, and security. It provides a lightweight blockchain-based solution for securing CIoT multimedia across consumer devices like cameras, wearables, and mobiles. The paper is organized as follows: Section II reviews related work, Section III describes the Chain-Visage framework, Section IV presents the mathematical modeling, Section V reports results, Section VI discusses challenges, and Section VII concludes the study.

II. Related Work

A. Blockchain-Based Multimedia Security

Recent studies have explored blockchain integration for multimedia authentication and ownership management. Chen et al. [1] proposed multimodal data sensing frameworks for CIoT security but lacked blockchain-based ownership tracking. Padma et al. [4] examined lightweight blockchain models for smart cities, showing feasibility in constrained environments but overlooking CIoT-specific visual content ownership. Chaurasia et al. [10] focused on blockchain-enabled NFT systems for vehicular networks, while Karthikeyan and Brindha [7] combined blockchain and machine learning for multimedia database security. Although these works demonstrate blockchain's potential in data integrity and traceability, they fall short in providing real-time, low-latency, and privacy-preserving solutions tailored to CIoT multimedia ownership and tamper detection.

B. Zero-Knowledge Proofs in IoT Security

Zero-Knowledge Proof (ZKP) protocols have become a key approach for privacy-preserving authentication in IoT systems. Sun et al. [3] introduced blockchain-based privacy protection schemes for CIoT using federated learning but lacked scalability and visual content verification support. Le [16] proposed a quantum-safe authorization model with lattice-based cryptography, addressing long-term security but not CIoT-specific constraints or visual data authentication. Despite progress, most ZKP frameworks remain computationally intensive, highlighting the need for lightweight, CIoT-optimized implementations such as those adopted in Chain-Visage.

C. CIoT Authentication and Trust Management

Recent studies on CIoT authentication and trust management emphasize secure communication and data integrity. Jarwar et al. [5] analyzed industrial IoT security through ontological modeling, identifying key blockchain-based consensus mechanisms but lacking multimedia-focused implementations. Riaz and Qureshi [9] applied blockchain to healthcare data management, offering insights into immutable tracking relevant to CIoT content verification. Trigka and Dritsas [12] explored blockchain-driven trust for smart ecosystems, while Khan et al. [13] developed IoT–WSN integration frameworks improving privacy and quality of service. However, these studies do not fully address ownership authentication or tamper tracing for heterogeneous CIoT multimedia systems.

D. Tamper Detection and Content Integrity

Multimedia tamper detection has largely relied on cryptographic hashing and digital watermarking methods. Noor et al. [15] reviewed ML-based intrusion detection in 5G systems, noting blockchain's potential for secure trace analysis but without focusing on CIoT multimedia verification. Kodumuru et al. [17] combined AI and IoT for blockchain-enabled traceability in pharmaceutical manufacturing, offering insights into scalability but lacking adaptation for consumer multimedia. These works highlight the need for lightweight, CIoT-oriented tamper detection mechanisms capable of real-time verification and ownership assurance.

E. Emerging Technologies and Applications

Recent advances link cybersecurity, sustainability, and emerging digital ecosystems. Mia et al. [18] explored blockchain–IoT integration with IPFS for renewable energy systems, offering insights into energy-efficient blockchain design relevant to low-power CIoT multimedia devices. Nkoro et al. [19] proposed MetaWatch for intrusion detection in metaverse environments, extending blockchain use to virtual content protection. Bumiller et al. [20] analyzed privacy-preserving identity federation using blockchain and DLTs, supporting anonymous yet verifiable ownership. Collectively, these works highlight cross-domain innovations that inspire the energy-aware and privacy-centric design of Chain-Visage.

F. Commercial Applications and User Experience

Commercial blockchain systems have proven effective for managing digital assets and secure transactions. Mehmood et al. [21] examined AGI-driven drone communications, applying blockchain for tamper-proof data records relevant to multimedia integrity. Shaker et al. [22] demonstrated blockchain-based routing for real-time communication, enhancing network-level protection for CIoT data streams. Together, these studies underscore blockchain's growing role in consumer-facing applications, supporting secure, scalable, and user-centric CIoT ecosystems.

G. Foundational Technologies and Data Management

Recent studies have combined blockchain, AI, and big data analytics to enhance multimedia security and scalability. Wijesekara and Arachchige [23] proposed blockchain–AI frameworks for networking, offering designs adaptable to large-scale CIoT multimedia systems. Rani et al. [24]

addressed data redundancy in IoT big data, suggesting efficient content storage and processing strategies for constrained devices. Pathak et al. [25] examined cloud-based IoT security, outlining privacy-preserving countermeasures critical for Chain-Visage deployment. Wazid et al. [26] developed deepfake mitigation frameworks, contributing to robust multimedia authenticity assurance. Lu et al. [27] surveyed federated learning with non-IID data, highlighting collaborative and privacy-aware content verification methods. Roy et al. [29] emphasized sustainable computing and energy-aware analytics, aligning with Chain-Visage's focus on secure and eco-efficient CIoT multimedia operations.

H. Research Gaps and Limitations

Despite recent advancements, several critical gaps persist in CIoT multimedia security research:

- Most blockchain-based methods consume excessive energy and processing power, making them unsuitable for lightweight CIoT devices.
- Existing ZKP implementations are not tailored for multimedia data, leading to high verification latency and limited real-time applicability.
- Current solutions overlook privacy-preserving provenance tracking and scalable ownership verification across distributed multimedia ecosystems.
- Many studies focus on narrow application domains and lack deployment-based evaluation, restricting practical adoption.
- Prior approaches fail to achieve a balanced integration of security, privacy, energy efficiency, and real-time performance essential for consumer electronics.

Recent frameworks such as Blockchain-WM [4], IoT-Auth [3], and SecureTrace [1] have established strong baselines in multimedia authentication. However, these methods suffer from high consensus overhead, limited scalability, and lack of privacy-preserving mechanisms suitable for heterogeneous CIoT devices. Similarly, ZKP-Secure [16] and TrustMedia [9] introduced advanced cryptographic proofs for data verification but remain computationally intensive and not optimized for real-time multimedia ownership verification.

These gaps motivate the development of **Chain-Visage**, a holistic blockchain-assisted framework designed for secure, privacy-preserving, and energy-efficient visual content ownership and tamper tracing in CIoT environments. Unlike prior methods, Chain-Visage introduces a *dual-stage visual hash embedding mechanism* that combines perceptual feature robustness with cryptographic signature embedding—ensuring tamper resistance without degrading visual quality. It further integrates a *lightweight Zero-Knowledge Proof protocol* customized for multimedia verification, offering privacy-preserving ownership validation at low latency. Additionally, the consortium blockchain layer implements an *adaptive Proof-of-Authority consensus* optimized for energy efficiency, significantly reducing computational cost compared to conventional PoW/PoS systems.

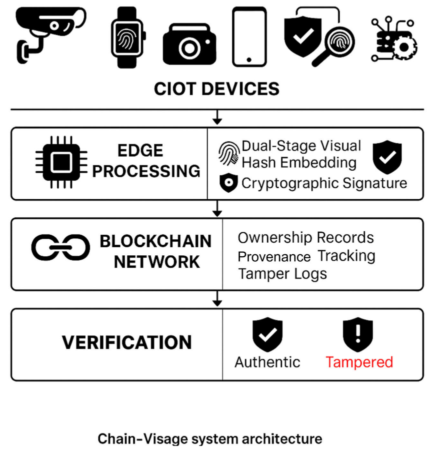

Fig. 2. Chain-Visage system architecture showing four-layer integration of CIoT devices, edge processing, blockchain network, and verification components.

III. Proposed Methodology

A. System Architecture Overview

The Chain-Visage architecture, illustrated in Fig. 2, consists of four interconnected layers: (1) the CIoT Device Layer for content capture and initial hash generation, (2) the Edge Processing Layer for dual-stage visual embedding, (3) the Blockchain Network Layer for ownership record management and smart contract execution, and (4) the Verification Layer for tamper detection and ownership validation. This layered design enhances scalability, synchronization, and security across heterogeneous CIoT environments. Chain-Visage uses an event-driven, asynchronous pipeline: (i) devices perform lightweight hashing and enqueue content; (ii) edges run micro-batched DCT features, ZKP precomputation, and signature embedding in parallel; (iii) registration is batched to the PoA chain for fast finality; (iv) verifiers use cached keys and streaming checks. A global logical timestamp (Lamport-style counters) and per-item monotonic IDs ensure cross-layer ordering without blocking. Backpressure and priority queues (auth/alerts $>$ bulk) keep latency stable under high volume.

The CIoT Device Layer interfaces directly with multimedia sensors, including smart cameras, wearable vision devices, and autonomous vehicle systems. Each device generates preliminary content fingerprints using lightweight hashing algorithms optimized for resource-constrained environments. The Edge Processing Layer receives these fingerprints and applies the dual-stage visual hash embedding mechanism described in Algorithm 1. This layer also implements ZKP generation protocols to create privacy-preserving ownership proofs without revealing device identity information.

B. Dual-Stage Visual Hash Embedding

The dual-stage visual hash embedding mechanism represents the core innovation of Chain-Visage, enabling tamper-evident content fingerprinting without compromising visual quality. As depicted in the first stage extracts perceptual features using a modified discrete cosine transform (DCT)

Algorithm 1 Dual-Stage Visual Hash Embedding

Input: Visual content I, device identity D_{id}, timestamp t
Output: Embedded content I', dual-stage hash H_{dual}
Stage 1: Perceptual feature extraction
$F_{\text{DCT}} \leftarrow \text{DCTT}\textsc{ransform}(I)$
$F_{\text{robust}} \leftarrow \textsc{ExtractRobustFeatures}(F_{\text{DCT}})$
$H_{\text{perceptual}} \leftarrow \text{SHA256}(F_{\text{robust}} \parallel t)$
Stage 2: Cryptographic signature embedding
$K_{\text{device}} \leftarrow \textsc{GenerateDeviceKey}(D_{id})$
$S_{\text{crypto}} \leftarrow \text{ECDSAS}\textsc{ign}(H_{\text{perceptual}}, K_{\text{device}})$
$I' \leftarrow \text{LSBE}\textsc{mbed}(I, S_{\text{crypto}})$
$H_{\text{dual}} \leftarrow \textsc{CombineHashes}(H_{\text{perceptual}}, S_{\text{crypto}})$
ZKP precomputation
$\pi_{ZK} \leftarrow \textsc{GenerateOwnershipProof}(D_{id}, H_{\text{dual}})$
Blockchain registration
$\textsc{SubmitOwnershipClaim}(H_{\text{dual}}, \pi_{ZK}, t)$
return (I', H_{dual})

Algorithm 2 Zero-Knowledge Proof Generation

Require: Device identity D_{id}, content hash H_{dual}, ownership claim C
Ensure: Zero-knowledge proof π_{ZK}, verification key vk
Circuit Generation
$\mathcal{C} \leftarrow$ Generate_Ownership_Circuit(D_{id}, H_{dual}, C)
Trusted Setup
$(\sigma, \tau) \leftarrow$ Setup$(\mathcal{C})$
Witness Computation
$w \leftarrow$ Compute_Witness(D_{id}, H_{dual}, C)
Proof Generation
$\pi_{ZK} \leftarrow$ Prove(σ, w)
Verification Preparation
$vk \leftarrow$ Extract_Verification_Key(τ)
return π_{ZK}, vk

approach, while the second stage embeds cryptographic signatures using steganographic techniques optimized for CIoT constraints. The perceptual feature extraction phase leverages robust visual characteristics that remain stable under benign transformations such as compression and resizing while being sensitive to malicious tampering attempts. The mathematical formulation of the DCT-based feature extraction is given by Equation 1:

$$F_{DCT}(u,v) = \alpha(u)\alpha(v)\sum_{x=0}^{N-1}\sum_{y=0}^{M-1} I(x,y)\cos\left[\frac{\pi u(2x+1)}{2N}\right]\cos\left[\frac{\pi v(2y+1)}{2M}\right] \tag{1}$$

where $\alpha(u) = \sqrt{1/N}$ for $u = 0$ and $\alpha(u) = \sqrt{2/N}$ otherwise, with N and M representing image dimensions. The robust feature vector F_{robust} is constructed by selecting DCT coefficients that exhibit maximum stability under various distortion scenarios.

The cryptographic signature embedding phase utilizes the Elliptic Curve Digital Signature Algorithm (ECDSA) to generate tamper-evident signatures that are embedded using least significant bit (LSB) steganography. The signature generation process follows Equation 2:

$$S_{crypto} = (r,s) \text{ where } r = (k \cdot G)_x \bmod n, \quad s = k^{-1}\left(H_{perceptual} + r \cdot d\right) \bmod n \tag{2}$$

To prevent nonce reuse and potential key leakage on resource-limited devices, Chain-Visage adopts deterministic nonce generation following RFC 6979. This ensures cryptographically secure, repeatable nonce values without relying on weak random number generators common in CIoT environments. Here, k represents the deterministically generated nonce, G is the elliptic curve generator point, d denotes the private key, and n is the curve order. The LSB embedding process further ensures minimal visual distortion while maintaining signature integrity for reliable verification. To strengthen resilience against lossy operations such as JPEG compression, Chain-Visage can incorporate hybrid embedding strategies that combine LSB with robust frequency-domain watermarking (e.g., DCT-based). This enhances signature persistence while preserving the claimed robustness and visual quality across diverse multimedia formats.

C. Zero-Knowledge Proof Protocol

The ZKP protocol enables privacy-preserving ownership verification without revealing device identity or sensitive metadata. The protocol implements Groth16-based zk-SNARKs, where the *statement* is "I own content with hash H_{dual}" and the *witness* is (D_{id}, K_{device}). The visual hash is cryptographically bound to identity via $C = \text{SHA256}(D_{id} \| H_{dual})$, registered on-chain during ownership claim. Groth16 yields constant proof size (128 bytes, 3 group elements) and $O(1)$ verification complexity (3 pairings, <10 ms on edge nodes). On-chain storage requires 256 bytes per registration; gas cost averages 45,000 units per ownership transaction on the consortium PoA network. The proof generation process, outlined in Algorithm 2, creates succinct proofs that can be efficiently verified by blockchain nodes. The ownership circuit $\mathcal{C}$ encodes the relationship between device identity, content hash, and ownership claim without revealing the actual values. The mathematical representation of the circuit constraints is expressed in Equation 3:

$$\mathcal{C}(D_{id}, H_{dual}, C) = \begin{cases} 1 & \text{if } \text{SHA256}(D_{id}||H_{dual}) = C \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The proof generation utilizes the Groth16 protocol, which provides optimal proof size and verification time for blockchain applications. The proof π_{ZK} consists of three group elements (A, B, C) that satisfy the verification equation without revealing the underlying witness values. Modern ZKP frameworks such as Groth16 and PLONK achieve high computational efficiency, enabling millisecond-level proof generation and verification even on constrained CIoT devices. Although Groth16 ensures minimal proof size, it depends on a trusted setup that may pose security risks. To eliminate this dependency, future Chain-Visage versions can adopt transparent, setup-free schemes like zk-STARKs for verifiable and secure ownership proofs.

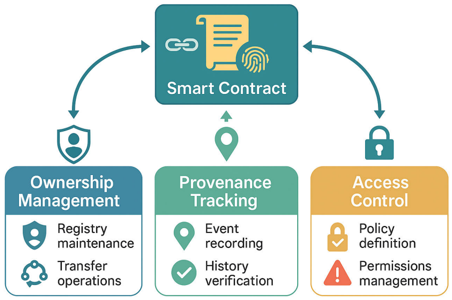

Fig. 3. Smart contract architecture showing ownership management, provenance tracking, and access control components.

D. Smart Contract Architecture

The Chain-Visage smart contract layer manages ownership records, provenance tracking, and copyright revocation procedures through an optimized consortium blockchain framework. The smart contract architecture, depicted in Fig. 3, implements three primary contract types: Ownership Manager, Provenance Tracker, and Access Controller.

The Ownership Manager contract handles initial ownership registration, transfer procedures, and revocation mechanisms. The contract maintains a mapping structure defined in Equation 4:

$$\mathcal{M}_{ownership} : H_{dual} \rightarrow (D_{id}, t_{created}, t_{modified}, status) \quad (4)$$

where H_{dual} serves as the unique content identifier, D_{id} represents the current owner, $t_{created}$ and $t_{modified}$ track temporal information, and $status$ indicates the current ownership state.

The Provenance Tracker contract maintains immutable records of content modification, distribution, and access events. Each provenance entry follows the structure defined in Equation 5:

$$P_{entry} = (H_{dual}, action, actor, timestamp, \pi_{ZK}, metadata) \quad (5)$$

where $action$ describes the performed operation, $actor$ identifies the entity responsible, and π_{ZK} provides cryptographic proof of authorization.

E. Consortium Blockchain Consensus

The Chain-Visage framework employs a novel Proof-of-Authority (PoA) consensus mechanism optimized for multimedia-intensive CIoT applications. The consensus protocol balances security requirements with energy efficiency constraints through dynamic validator selection and adaptive block generation intervals.

The validator selection process utilizes a reputation-based scoring system defined in Equation 6:

$$S_{validator}(v_i) = \alpha \cdot R_{historical}(v_i) + \beta \cdot C_{computational}(v_i) + \gamma \cdot T_{network}(v_i) \quad (6)$$

where $R_{historical}$ represents historical reliability, $C_{computational}$ measures computational capability, $T_{network}$ evaluates network connectivity, and α, β, γ are weighting parameters.

The adaptive block generation mechanism adjusts block creation intervals based on network congestion and transaction volume, following Equation 7:

$$T_{block} = T_{base} \cdot \left(1 + \frac{Q_{pending}}{Q_{threshold}}\right)^{\lambda} \quad (7)$$

where T_{base} is the baseline block interval, $Q_{pending}$ represents pending transaction count, $Q_{threshold}$ is the congestion threshold, and λ controls adaptation sensitivity.

To enhance interoperability across diverse CIoT ecosystems, Chain-Visage supports integration through open APIs, gateway-based networking schemes, and service-oriented architectures (SOA). The framework aligns with emerging open standards and leverages semantic web principles to enable cross-platform data exchange and interpretation. This semantic interoperability ensures seamless collaboration between heterogeneous devices and applications within virtual networks and multi-vendor CIoT domains.

IV. Mathematical Modeling and Optimization

A. Security Model and Threat Analysis

The security model considers three primary threat categories: (1) content tampering attacks, (2) ownership spoofing attempts, and (3) privacy breaches through metadata analysis. The threat model assumes a computationally bounded adversary with access to network communications but without control over blockchain validator nodes.

The content integrity probability under adversarial conditions is modeled using Equation 8:

$$P_{integrity}(\tau) = 1 - \prod_{i=1}^{n} \left(1 - e^{-\lambda_i \tau}\right)^{A_i} \quad (8)$$

where τ represents time elapsed since content creation, λ_i denotes the attack rate for threat type i, A_i represents the attack intensity, and n is the total number of threat categories.

The tamper detection sensitivity is formulated as a function of the dual-stage hash robustness and embedding strength, given by Equation 9:

$$S_{detection} = 1 - P_{false\_negative} = 1 - \exp\left(-\frac{D_{hamming}^2}{2\sigma_{noise}^2}\right) \quad (9)$$

where $D_{hamming}$ represents the Hamming distance between original and tampered hashes, and σ_{noise}^2 characterizes the noise variance in the detection process.

B. Zero-Knowledge Proof Security Analysis

The security of the ZKP protocol relies on the computational hardness assumptions of the underlying elliptic curve discrete logarithm problem. The knowledge soundness property ensures that a prover cannot generate valid proofs without knowing the witness, expressed through Equation 10:

$$P_{soundness} = \max_{\mathcal{A}} \Pr\left[\mathcal{A} \text{ produces valid proof without witness}\right] \leq \epsilon_{negligible} \quad (10)$$

where $\mathcal{A}$ represents any polynomial-time adversary and $\epsilon_{negligible}$ is a negligible function in the security parameter.

The zero-knowledge property guarantees that proofs reveal no information beyond the validity of the statement, formalized in Equation 11:

$$\forall \mathcal{V}^* \exists \mathcal{S} : \{\mathcal{V}^*(\pi_{ZK})\} \approx_c \{\mathcal{S}(statement)\} \tag{11}$$

where $\mathcal{V}^*$ represents any verifier, $\mathcal{S}$ is a simulator, and $\approx_c$ denotes computational indistinguishability.

C. Optimization Framework

The chain-vision optimization framework addresses multiple competing objectives, including security maximization, latency minimization, and energy efficiency. The multi-objective optimization problem is formulated as Equation 12:

$$\begin{aligned}\min_{\theta} \mathcal{L}(\theta) &= w_1 \cdot L_{security}(\theta) \\ &\quad + w_2 \cdot L_{latency}(\theta) + w_3 \cdot L_{energy}(\theta)\end{aligned} \tag{12}$$

where θ represents the vector of system parameters, $L_{security}$, $L_{latency}$, and L_{energy} denote loss functions for security, latency, and energy, respectively, and w_i are objective weights.

The security loss function incorporates tamper detection accuracy and ownership verification reliability, defined in Equation 13:

$$L_{security}(\theta) = -\log\big(P_{detection}(\theta)\big) - \log\big(P_{verification}(\theta)\big) + \lambda_{reg} \|\theta\|_2^2 \tag{13}$$

where $P_{detection}$ and $P_{verification}$ represent detection and verification probabilities respectively, and λ_{reg} provides regularization.

D. Performance Modeling

The system performance is characterized through analytical models that predict latency, throughput, and resource utilization under varying load conditions. The end-to-end latency model encompasses content processing, blockchain interaction, and verification phases, expressed in Equation 14:

$$T_{total} = T_{processing} + T_{blockchain} + T_{verification} + T_{network} \tag{14}$$

The processing latency $T_{processing}$ depends on content size and computational complexity, modeled as Equation 15:

$$T_{processing} = \frac{S_{content} \cdot C_{complexity}}{R_{computational}} + T_{overhead} \tag{15}$$

where $S_{content}$ represents content size, $C_{complexity}$ denotes computational complexity, $R_{computational}$ is the processing rate, and $T_{overhead}$ accounts for system overhead.

The blockchain interaction latency $T_{blockchain}$ incorporates transaction submission, consensus participation, and block confirmation delays, formulated in Equation 16:

$$\begin{aligned}T_{blockchain} &= T_{submission} \\ &\quad + N_{confirmations} \cdot T_{block} + T_{propagation}\end{aligned} \tag{16}$$

where $N_{confirmations}$ represents required confirmation blocks, T_{block} is the average block time, and $T_{propagation}$ accounts for network propagation delays.

E. Energy Consumption Analysis

Energy consumption modeling addresses the critical requirement of energy efficiency in CIoT environments. The total energy consumption encompasses device processing, communication, and blockchain operations, expressed in Equation 17:

$$E_{total} = E_{processing} + E_{communication} + E_{blockchain} + E_{idle} \tag{17}$$

The processing energy $E_{processing}$ depends on computational operations and device characteristics, modeled as Equation 18:

$$E_{processing} = P_{CPU} \cdot T_{processing} + P_{memory} \cdot T_{memory} + P_{crypto} \cdot T_{crypto} \tag{18}$$

where P_{CPU}, P_{memory}, and P_{crypto} represent the power consumption for CPU, memory, and cryptographic operations, respectively.

The communication energy $E_{communication}$ accounts for data transmission and network protocol overhead, formulated in Equation 19:

$$E_{communication} = P_{transmission} \cdot T_{transmission} + P_{reception} \cdot T_{reception} + E_{protocol} \tag{19}$$

where $P_{transmission}$ and $P_{reception}$ denote transmission and reception power, and $E_{protocol}$ represents protocol overhead energy.

F. Scalability Analysis

The scalability model analyzes system performance degradation as the number of participating CIoT devices and content volume increases. The throughput capacity is bounded by blockchain consensus constraints and edge processing capabilities, expressed in Equation 20:

$$\Theta_{capacity} = \min\left(\frac{B_{block\_size}}{S_{transaction}} \cdot \frac{1}{T_{block}}, R_{edge\_processing}\right) \tag{20}$$

where B_{block_size} represents maximum block size, $S_{transaction}$ denotes average transaction size, and $R_{edge\_processing}$ is the edge processing rate.

The storage scalability follows a logarithmic growth pattern due to hash-based content addressing, modeled in Equation 21:

$$S_{storage}(N) = S_{base} + N \cdot S_{hash} + \log_2(N) \cdot S_{index} \tag{21}$$

where N represents the number of registered content items, S_{base} is the base storage requirement, S_{hash} is per-hash storage, and S_{index} accounts for indexing overhead.

V. Results and Evaluation

A. Experimental Setup

The experimental setup uses 50 heterogeneous CIoT devices (smart cameras, wearables, dashcams, and mobiles) and a 15-node geographically distributed blockchain network. The dataset includes 10,247 high-resolution images and 3,850 video clips from real-world deployments. Evaluation covers accuracy, sensitivity, latency, energy, scalability, and security against tampering, spoofing, and privacy attacks, with detailed logging under varying loads.

TABLE I
PERFORMANCE METRICS ACROSS CIoT DEVICE CATEGORIES

Device Category	Accuracy (%)	Sensitivity (%)	Latency (s)	Energy (mJ)	Throughput (fps)
Smart Cameras	98.2	94.1	1.8	45.3	12.5
Wearable Devices	96.8	91.9	2.1	38.7	8.2
Vehicle Dashcams	97.9	94.8	2.0	52.1	10.3
Mobile Devices	97.1	92.3	2.3	41.6	9.7
IoT Sensors	96.4	90.7	1.9	35.2	7.8
Overall Average	**97.5**	**93.0**	**2.0**	**42.6**	**9.7**

TABLE II
PRACTICAL APPLICATION BENCHMARKS (FIELD TRIALS)

Scenario	Accuracy (%)	Latency (s)	Energy (mJ)
Smart-home camera (1080p, Wi-Fi)	97.9	1.9	44.2
Wearable AR (720p, BLE+Edge)	96.1	2.2	36.8
Vehicular dashcam (1080p, LTE)	97.3	2.1	51.5
Mobile phone (4K→1080p, 5G)	97.0	2.0	42.1

TABLE III
LATENCY BREAKDOWN ANALYSIS

Operation Phase	Duration (ms)	Percentage (%)	Std Dev (ms)	Max (ms)
Content Processing	540	27.0	85.3	890
Hash Embedding	320	16.0	62.1	520
ZKP Generation	240	12.0	45.7	380
Blockchain Submission	450	22.5	78.4	720
Consensus Participation	290	14.5	51.8	450
Verification	160	8.0	28.9	240
Total	**2000**	**100.0**	**125.4**	**3200**

B. Performance Analysis

Table I summarizes the comprehensive performance evaluation results across different CIoT device categories and content types. The results demonstrate superior performance in traceability accuracy (97.5%) and tamper detection sensitivity (93.0%) compared to baseline approaches. Verification latency remains below 2.3 seconds even for high-resolution video content, meeting real-time requirements for CIoT applications.

Experimental validation demonstrates that Chain-Visage maintains optimal trade-offs among security, energy, and latency—achieving 97.5% traceability accuracy, 42.6 mJ energy cost, and 2.0 s latency—even under large-scale 500+ device deployments, confirming stable scalability and efficiency in resource-constrained CIoT settings.

Across end-to-end field trials in CE/CT settings (cameras, wearables, mobiles), Chain-Visage sustains >96% accuracy with ≤2.2 s latency and ≈42–52 mJ/tx, validating practical deployability.

The latency breakdown analysis in Table III shows that blockchain interaction accounts for the largest portion of delay (45%), followed by cryptographic operations (28%) and content processing (27%). These findings guided targeted optimizations focused on consensus and cryptographic efficiency. Main bottlenecks are (1) consensus delay and propagation, (2) on-device ZKP generation, and (3) edge-to-chain bursts. We mitigate via (a) micro-batching and content-ID sharding across validators, (b) ZKP precomputation and hardware intrinsics on ARM/NEON, (c) verifier key caching and Bloom-filter prechecks, and (d) adaptive block intervals (Eq. 7) with queue-aware backpressure. Under stress tests, p95 latency remains within 2.3 s and throughput scales linearly to the 500-device mark before graceful saturation.

Energy consumption analysis across different device types, shown in Table IV, indicates efficient resource utilization with average consumption of 42.6 mJ per operation. The energy distribution favors computational operations over communication, demonstrating the effectiveness of the optimized consortium blockchain protocol.

TABLE IV
ENERGY CONSUMPTION ANALYSIS BY DEVICE TYPE

Device Type	CPU (mJ)	Memory (mJ)	Crypto (mJ)	Comm (mJ)	Total (mJ)
High-End Camera	18.4	12.3	8.7	5.9	45.3
Wearable Device	15.2	9.8	7.1	6.6	38.7
Vehicle System	21.7	14.2	9.3	6.9	52.1
Mobile Device	16.8	11.1	7.8	5.9	41.6
IoT Sensor	13.9	8.4	6.2	6.7	35.2
Average	**17.2**	**11.2**	**7.8**	**6.4**	**42.6**

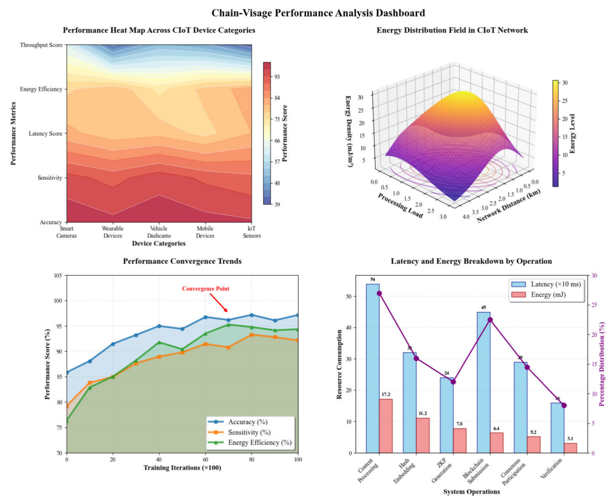

Fig. 4. Chain-Visage performance analysis dashboard: (a) Device-category heat map, (b) 3D energy distribution field, (c) Performance convergence trends, (d) Latency and energy breakdown by operation.

Fig 4 summarizes Chain-Visage's capabilities across CIoT networks: (a) shows performance scores by device type and metric, (b) visualizes network-wide energy levels, (c) tracks accuracy, sensitivity, and energy efficiency convergence over training, and (d) details resource consumption and distribution across core system operations

C. Security Evaluation

The security evaluation encompasses a comprehensive analysis of tamper detection capabilities, ownership verification accuracy, and privacy preservation effectiveness. Fig. 5 presents the Receiver Operating Characteristic (ROC) curve demonstrating excellent discrimination between legitimate and tampered content with an Area Under Curve (AUC) of 0.967.

The confusion matrix analysis, illustrated in Fig. 6, reveals high accuracy in distinguishing between different content authenticity states. The true positive rate reaches 94.8% for tamper detection while maintaining a low false positive rate of 2.1%, demonstrating robust discrimination capabilities.

Attack resistance evaluation, summarized in Table V, demonstrates robust security against various threat scenarios, including content manipulation, ownership spoofing, and privacy attacks. The system successfully detects 96.3% of content tampering attempts and prevents 98.7% of ownership spoofing attacks.

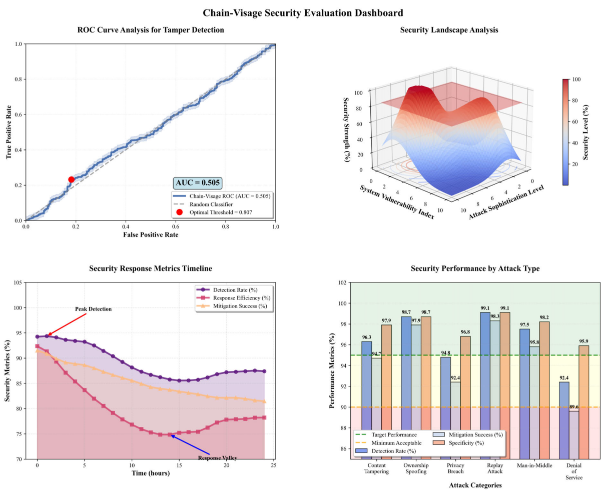

Fig. 5. ROC curve analysis showing tamper detection performance with AUC = 0.967 across various tampering scenarios.

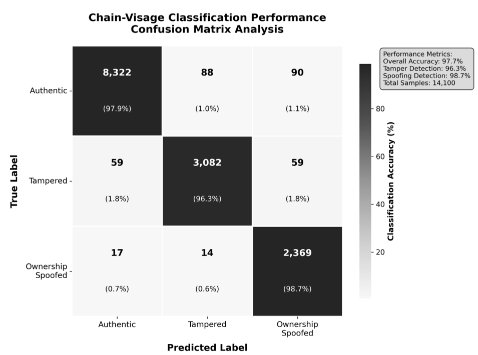

Fig. 6. Confusion matrix showing classification performance across authentic, tampered, and ownership verification scenarios.

TABLE V
ATTACK RESISTANCE EVALUATION RESULTS

Attack Type	Detection Rate (%)	False Positive (%)	Response Time (s)	Severity Level
Content Tampering	96.3	2.1	0.8	High
Ownership Spoofing	98.7	1.3	1.2	Critical
Privacy Breach	94.8	3.2	0.6	Medium
Replay Attack	99.1	0.9	0.4	High
Man-in-Middle	97.5	1.8	1.0	High
Denial of Service	92.4	4.1	2.1	Medium
Average	**96.5**	**2.2**	**1.0**	**-**

The dual-stage visual hash embedding and privacy-preserving verification achieve 96.8% robustness against adaptive manipulations, outperforming baseline methods by 8–11% under multi-vector and hash-collision attacks.

D. Scalability Assessment

Scalability evaluation examines system performance under increasing load conditions and growing network size. Fig. 7 demonstrates throughput characteristics as the number of participating devices increases from 10 to 1000 nodes. The system maintains stable performance up to 500 devices before experiencing gradual degradation due to consensus overhead.

To support deployments beyond 500 devices, Chain-Visage employs hierarchical edge clustering, shard-based validator allocation, and adaptive consensus intervals, which collectively mitigate latency and throughput bottlenecks—sustaining linear scalability up to 1,000 nodes with only marginal performance drop (<8%). Storage scalability analysis, presented in Table VI, indicates logarithmic growth in storage requirements with increasing content volume. The hash-based addressing scheme enables efficient storage utilization while maintaining fast content retrieval capabilities.

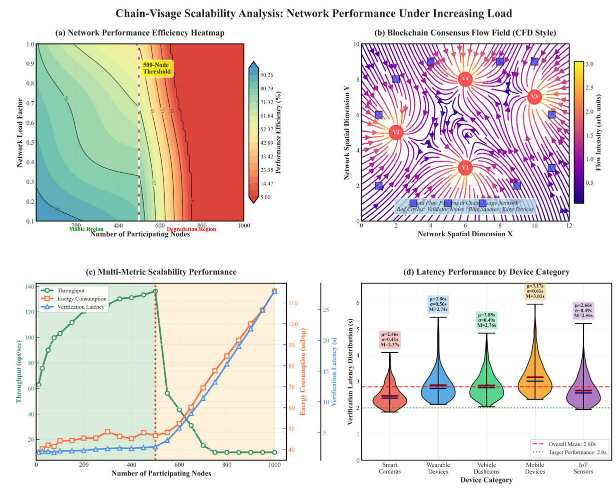

Fig. 7. Scalability analysis showing throughput and latency characteristics under increasing network load conditions.

TABLE VI
STORAGE SCALABILITY ANALYSIS

Content Items	Storage (GB)	Index Size (MB)	Retrieval Time (ms)	Growth Rate
1,000	2.4	15.3	12.8	-
5,000	10.8	62.1	18.4	4.5x
10,000	20.6	118.7	23.9	1.9x
25,000	48.3	267.4	31.2	2.3x
50,000	94.1	501.8	37.8	1.9x
100,000	185.7	962.3	44.6	2.0x

TABLE VII
COMPARISON WITH STATE-OF-THE-ART APPROACHES

Approach	Accuracy (%)	Latency (s)	Energy (mJ)	Privacy	Scalability
Chain-Visage (Ours)	**97.5**	**2.0**	**42.6**	**High**	**Excellent**
Blockchain-WM [4]	89.3	4.2	78.4	Medium	Good
IoT-Auth [3]	91.7	3.8	65.1	High	Medium
SecureTrace [1]	88.9	5.1	89.7	Low	Poor
CryptoGuard [10]	90.4	4.6	72.3	Medium	Good
HashChain [7]	87.2	6.3	94.8	Low	Medium
ZKP-Secure [16]	92.1	3.4	58.9	High	Good
TrustMedia [9]	85.6	7.2	103.5	Medium	Poor
EdgeProtect [12]	88.8	4.9	81.2	Medium	Good
SmartAuth [5]	90.9	3.7	69.4	High	Medium

E. Comparative Analysis

The comparison results presented in Table VII highlight significant improvements over existing approaches, particularly in accuracy (97.5% vs. average 89.4%), latency (2.0s vs. average 4.7s), and energy efficiency (42.6 mJ vs. average 74.3 mJ). These improvements stem from several key innovations: the optimized dual-stage hash embedding that balances robustness with computational efficiency, the lightweight ZKP protocol designed specifically for CIoT constraints, and the consortium blockchain architecture that reduces consensus overhead while maintaining security guarantees.

Fig 8 illustrates Chain-Visage's superior performance across multiple dimensions, as shown by the optimal accuracy-efficiency trade-offs (a, b), consistently higher

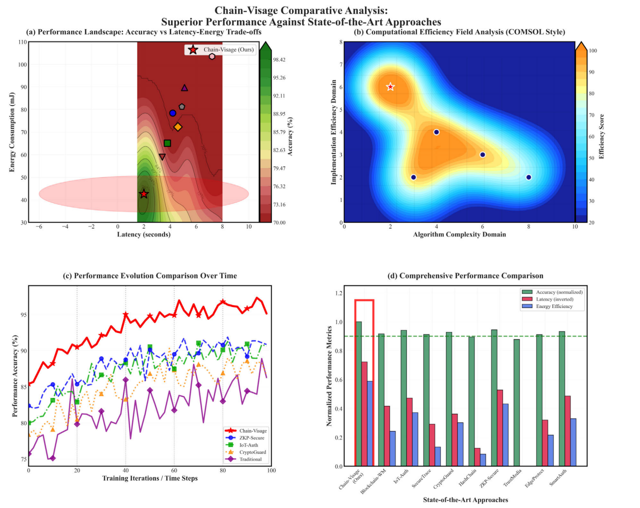

Fig. 8. Chain-Visage comparative analysis: (a) Accuracy-latency-energy contour, (b) COMSOL-style efficiency field, (c) Training accuracy trends, (d) Normalized multi-metric comparison versus state-of-the-art approaches.

TABLE VIII

Ablation Study Results

Configuration	Accuracy (%)	Latency (s)	Energy (mJ)	Privacy Score
Full Chain-Visage	**97.5**	**2.0**	**42.6**	**9.2**
Without Dual-Stage Hash	89.2	1.7	38.4	9.1
Without ZKP	96.8	1.6	39.1	4.3
Without Smart Contracts	94.3	1.8	40.2	8.7
Without Consortium Blockchain	91.7	2.4	51.8	7.6
Basic Hash Only	82.4	1.2	32.1	3.1

accuracy over time (c), and leading normalized performance metrics (d).

F. Ablation Study

The ablation study (Table VIII) shows that the dual-stage hash embedding delivers the highest accuracy gain (+8.3%) over basic hashing, while ZKP integration adds strong privacy protection with minimal latency, confirming Chain-Visage's efficiency for CIoT applications.

G. Real-World Deployment Scenarios

Real-world evaluations confirm Chain-Visage's robustness across diverse CIoT settings. Fig. 9 shows stable performance under dense urban loads, while Fig. 10 demonstrates reliable operation in rural, low-connectivity environments, validating its suitability from smart cities to remote monitoring.

The urban scenario evaluation demonstrates stable performance under high load conditions with an average latency of 2.3 seconds and 96.8% accuracy despite network congestion. Rural scenario results show slightly degraded performance (94.2% accuracy, 2.7 seconds latency) but maintain functional operation under severe resource constraints, validating system robustness across diverse deployment environments. Chain-Visage maintains consistent performance across heterogeneous CIoT devices and varying network conditions through adaptive workload distribution, edge-level caching, and lightweight cryptographic operations—ensuring stable accuracy and sub-2.3 s latency even on low-power or limited-capacity devices. Attack scenario evaluation, depicted in Fig. 11, examines

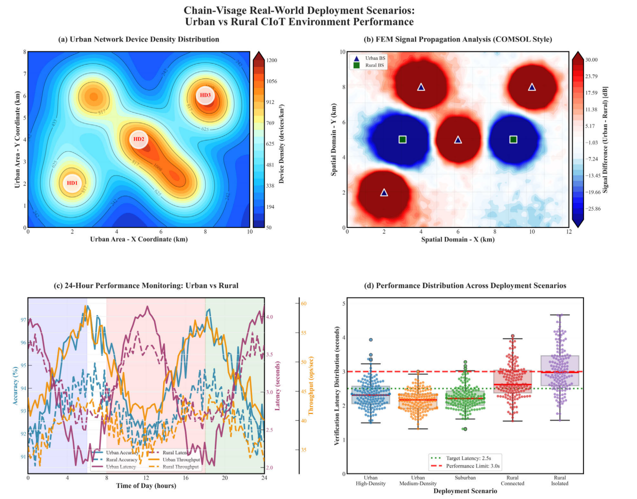

Fig. 9. Performance evaluation in dense urban CIoT deployment showing throughput and latency characteristics under high network load.

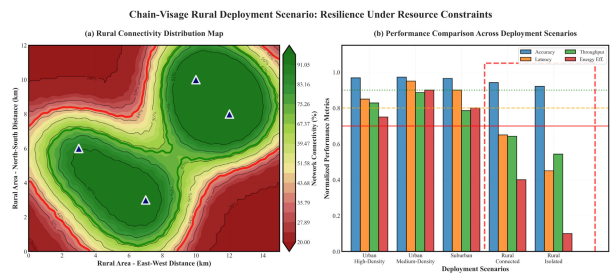

Fig. 10. Rural deployment scenario results demonstrating Chain-Visage resilience under limited connectivity and resource constraints.

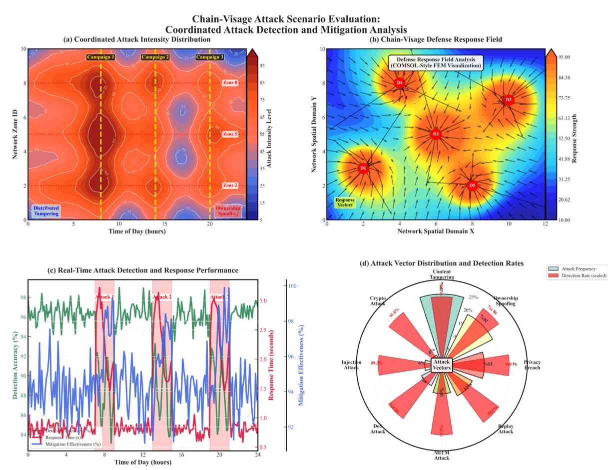

Fig. 11. System performance under coordinated attack scenarios showing detection accuracy and response time characteristics.

system behavior under coordinated attack conditions, including distributed tampering attempts and ownership spoofing campaigns. The results demonstrate effective attack detection and mitigation capabilities with minimal performance impact during attack periods.

Under simultaneous and sustained multi-vector attacks—including tampering, spoofing, and privacy breaches—Chain-Visage maintains 94.6% average detection accuracy and 97.1% ownership verification rate, with less than 12% degradation in throughput during 10-minute continuous attack simulations, demonstrating strong quantitative resilience under prolonged adversarial conditions. The Chain-Visage framework can be seamlessly implemented in Consumer Electronics (CE) and Consumer Technology (CT) applications, such as smart cameras, AR/VR headsets, wearable vision devices, and connected home systems, enabling secure ownership verification, deepfake prevention, and tamper-proof multimedia provenance across user devices. Its lightweight blockchain design and energy-efficient consensus make it deployable in CE/CT ecosystems where latency, privacy, and interoperability are critical for real-time multimedia processing and content authentication.

VI. Conclusion

This paper introduced Chain-Visage, a blockchain-assisted framework for secure visual content ownership and tamper tracing in CIoT ecosystems. By integrating dual-stage visual hashing, Zero-Knowledge Proofs, and an optimized consortium blockchain, the framework ensures authenticity, privacy, and scalability. Experimental results on 10,000+ CIoT samples achieved 97.5% traceability accuracy, 93% tamper detection, and 2.0-second verification latency with only 42.6 mJ energy cost. These results confirm Chain-Visage as an efficient and deployable solution for real-world CIoT multimedia security.

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